

6.104 · Software Design · MIT · Fall 2026

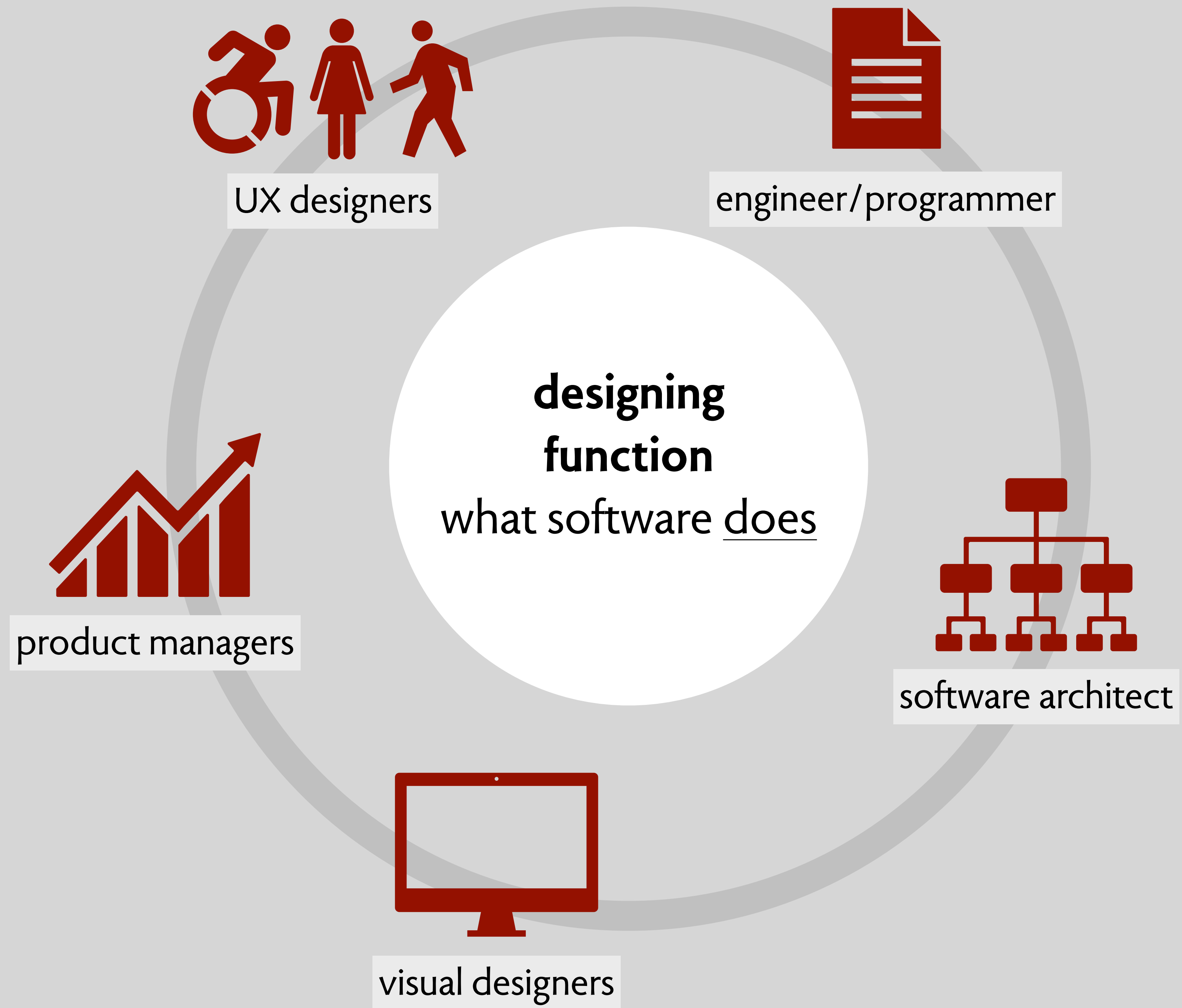
# intro to concept design

Daniel Jackson

what you'll learn today

what a concept is  
and how some companies  
have made their success on just one

what is software  
design about?





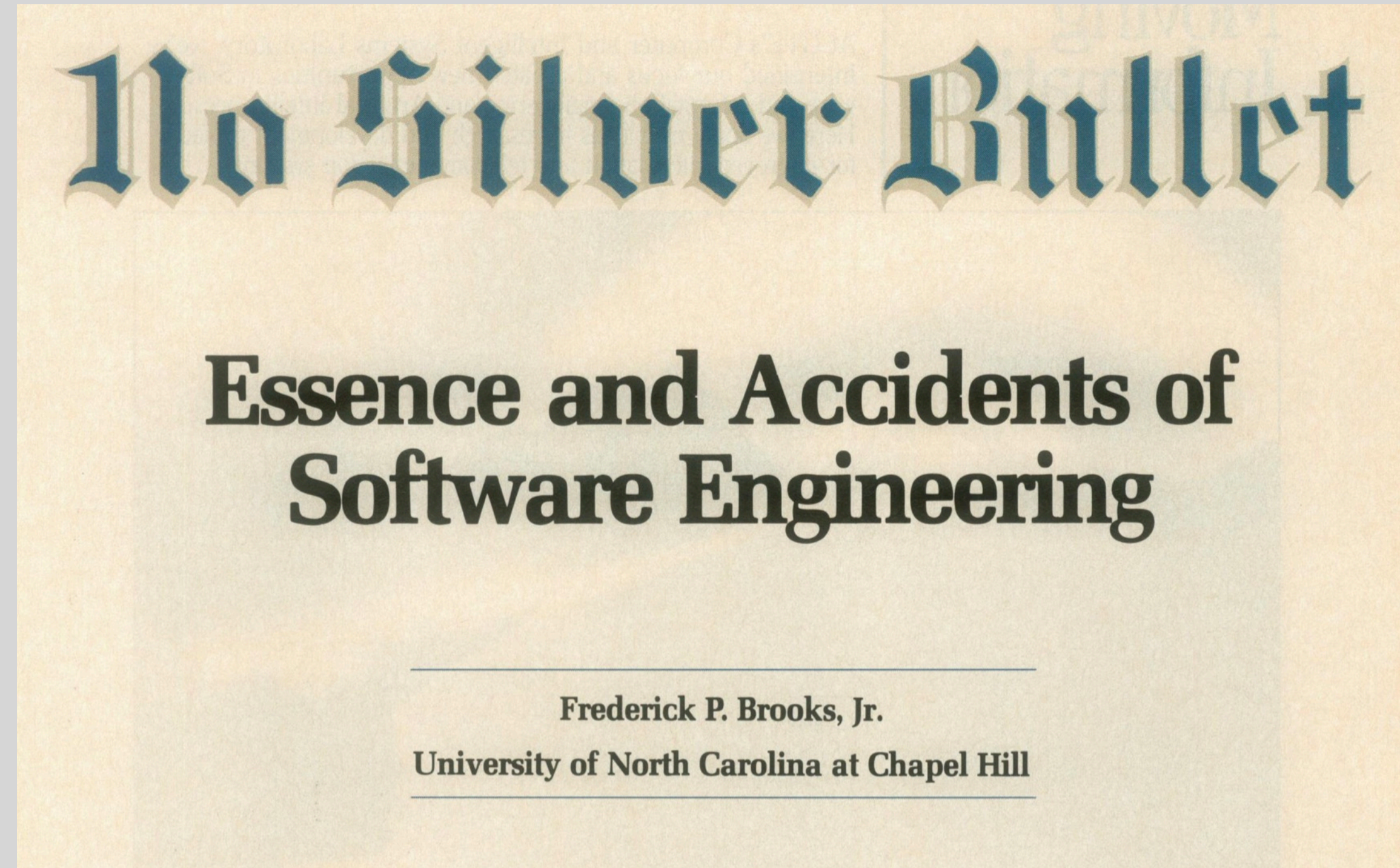
When you go to design a house you talk to an architect first, not an engineer. Why is this?

Because the criteria for what makes a good building fall outside the domain of engineering.

Similarly, in computer programs, **the selection of the various components** must be driven by the conditions of use.

How is this to be done? **By software designers.**

what AI cannot do



“The hardest single part of building a software system is deciding precisely what to build.”

# seeking a language of design

We live at the level of our language.

Whatever we can articulate  
we can imagine or understand or explore.

—Ellen Gilchrist

introducing  
concepts

# what's a concept?

a concept is a **realized** **course of action**

## **Inviting: the realization**

The host keeps the list of of guests,  
and records their responses.

a concept can be  
realized in any form,  
including software

## **Inviting: the course of action**

A host makes a guest list for an  
occasion, and then invites each guest.  
Each guest can accept or decline the  
invitation, until some time after which  
responses are no longer accepted.

# three key elements

*the name of the concept*

**concept** Inviting

*what the concept is for*

**purpose** ensure mutual consent for participation in some engagement

*how the concept works*

**principle** A host makes a guest list for an occasion, and then invites each guest. Each guest can accept or decline the invitation, until some time after which responses are no longer accepted.



Michael Polanyi (1891-1976)  
the inventor of the  
operational principle

# a different concept

*the name of the concept*

**concept** Reserving

*what the concept is for*

**purpose** secure future access to a scarce resource

*how the concept works*

**principle** A host makes some resources available for some occasion. A participant can reserve a resource and then claim it at the occasion.

# three more concepts

**concept** Upvoting

**purpose** rank items by popularity

**principle** Voters can upvote or downvote items that they approve or disapprove of, and then the items are ranked by their votes.

**concept** Rating

**purpose** share perceived quality of offerings as numbers to allow easy comparisons between them

**principle** Raters rate offerings on some numeric scale, leading to aggregate ratings that can be compared.

**concept** Escalating

**purpose** separate handling of issues requiring special treatment

**principle** An agent can escalate an issue, and later an agent can resolve an escalated issue.

should this talk about assigning the escalated issue? no, that's part of the Assigning concept.

what bad situations do these concepts prevent?

# including bad situations in the purpose

## concept Inviting

**purpose** ensure mutual consent for participation in some engagement;  
**prevents unwanted participation.**

**principle** A host makes a guest list for an occasion, and then invites each guest. Each guest can accept or decline the invitation, until some time after which responses are no longer accepted.

## concept Upvoting

**purpose** rank items by popularity;  
**prevents being overwhelmed with low quality items.**

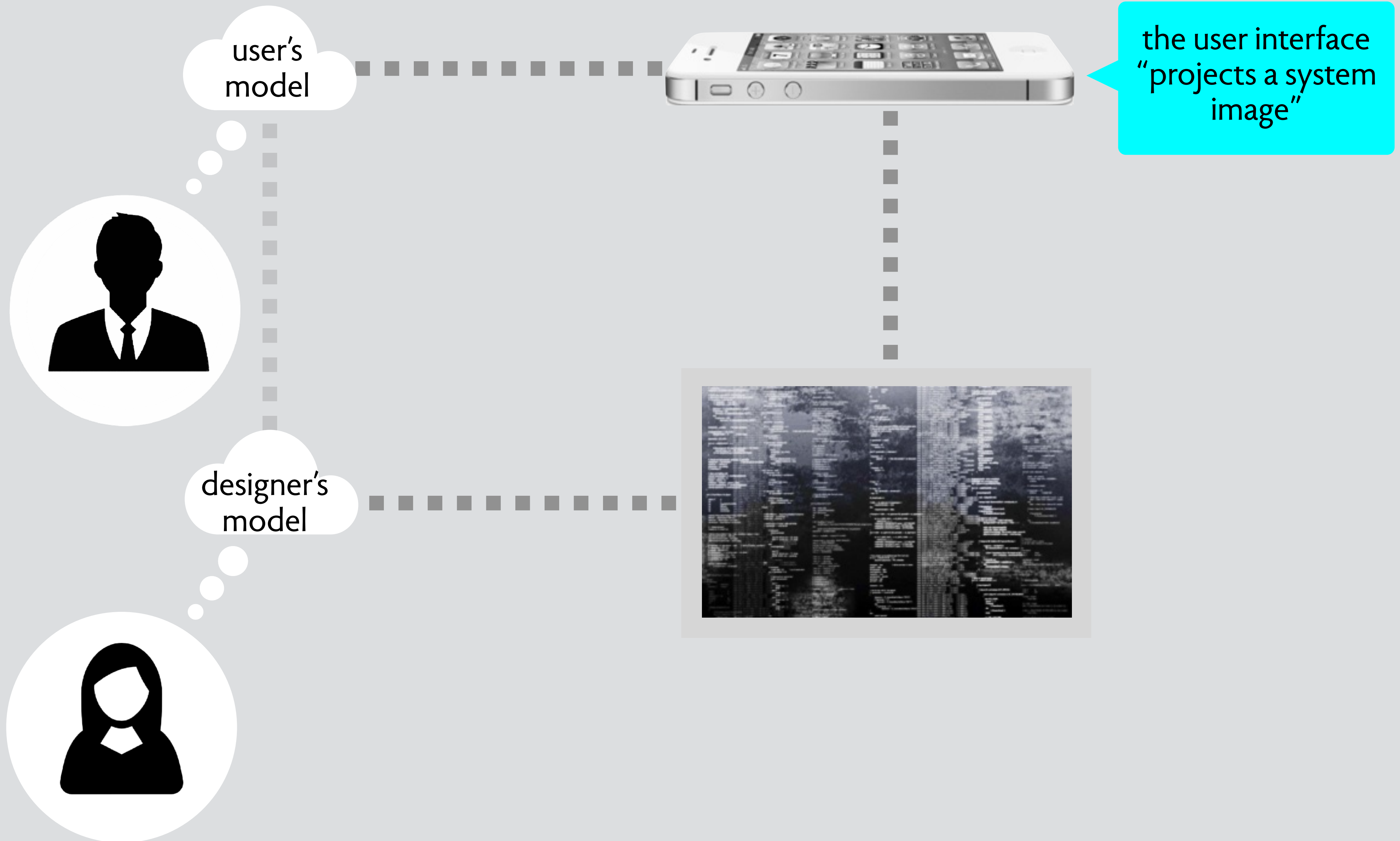
**principle** Voters can upvote or downvote items that they approve or disapprove of, and then the items are ranked by their votes.

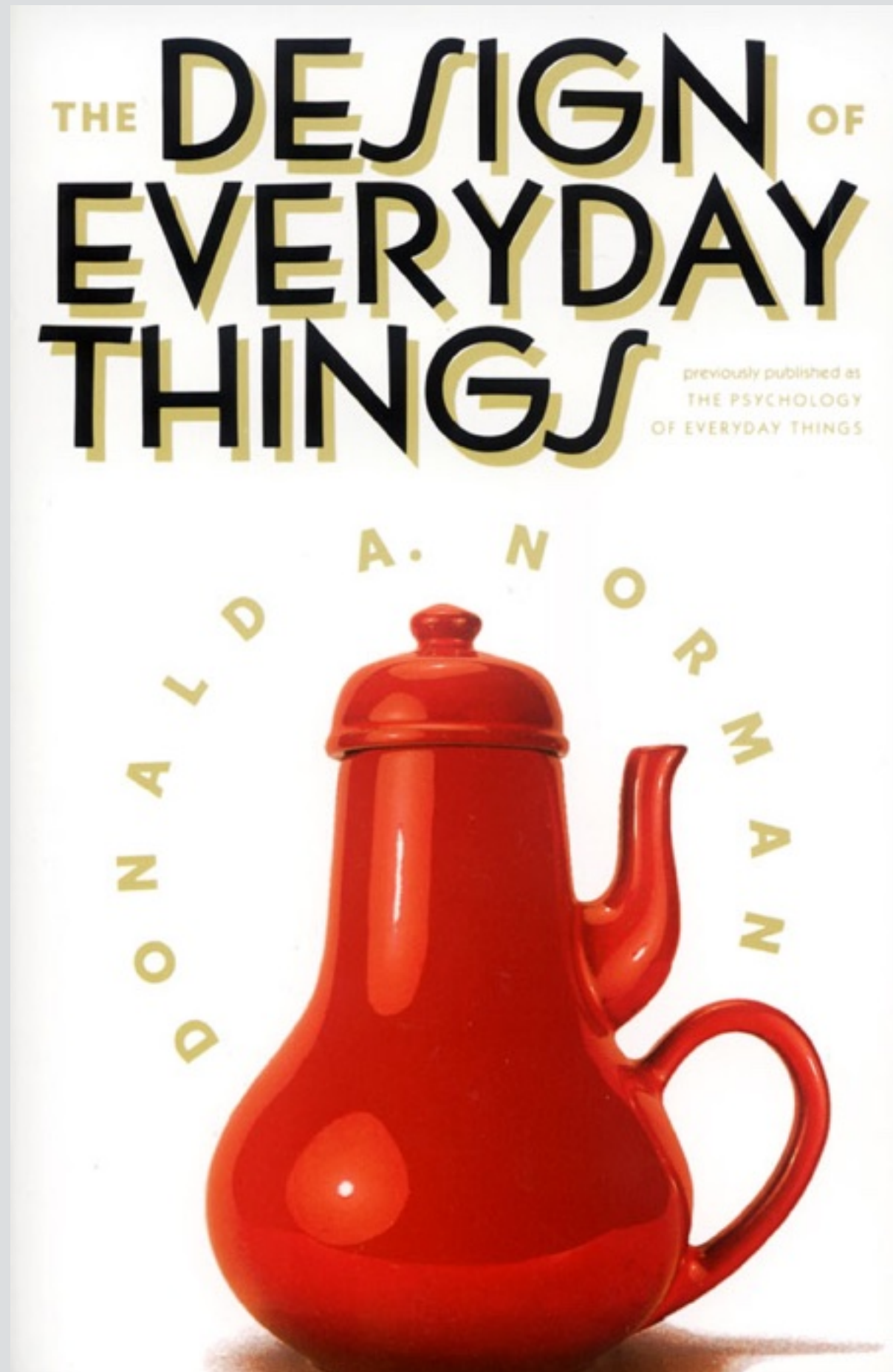
## concept Rating

**purpose** share perceived quality of offerings as numbers to allow easy comparisons between them  
**prevents inability to select between similar offerings**

**principle** Raters rate offerings on some numeric scale, leading to aggregate ratings that can be compared.

concepts &  
conceptual models





1988



Donald Norman

When the designers fail to provide a conceptual model, we will be forced to make up our own, and the ones we make up are apt to be wrong. **Conceptual models are critical to good design.** (2013 edition)

# what are the concepts in conceptual models?

## **how people talk**

“the concept of a restaurant reservation”

“the concept of a friend in a Facebook”

“the concept of a password”

## **the conceptual model is about behavior**

Reserving, Friending, PasswordAuthenticating

## **the object is not the concept!**

a restaurant reservation is a small part of the Reserving concept

the concept is the course of action (allocate, reserve, claim)

and about relationships (reservation is for this resource by this user)

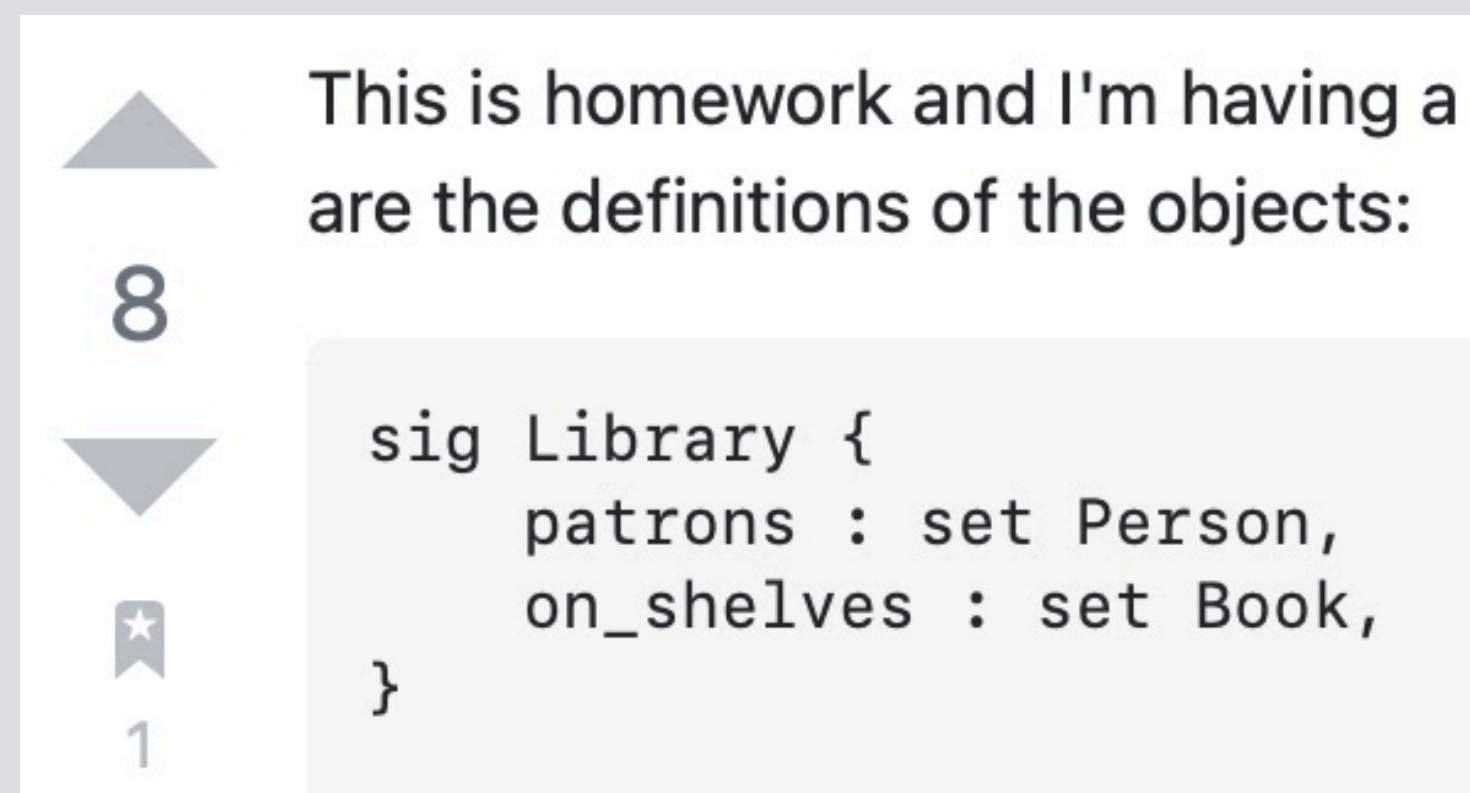
recognizing  
concepts

# similar UIs, very different concepts

**concept** Upvoting

**purpose** rank items by popularity

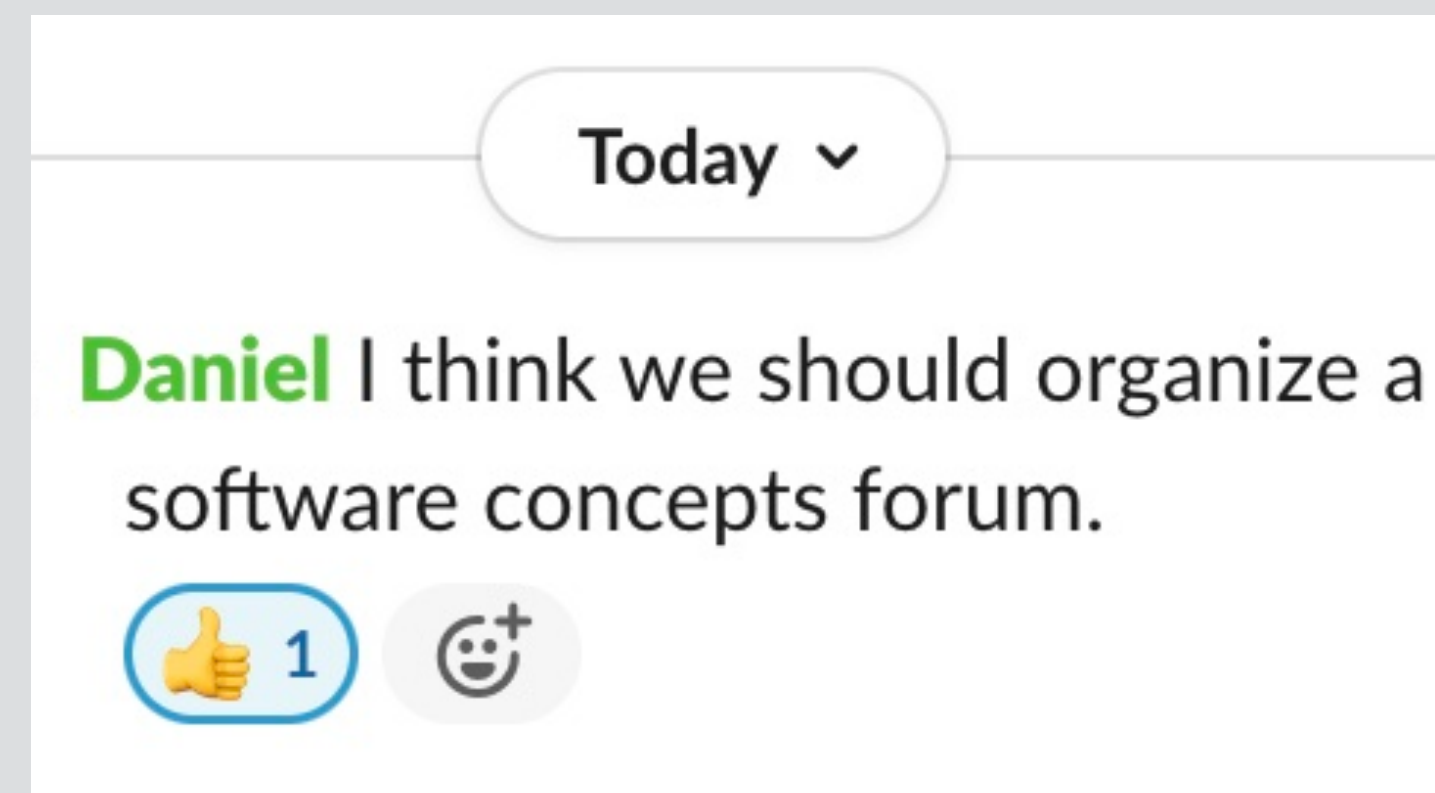
**principle** after series of upvotes of items, the items are ranked by their number of upvotes



**concept** Reacting

**purpose** share reactions to item

**principle** after users react to an item, the reactions are displayed (often in aggregated form)



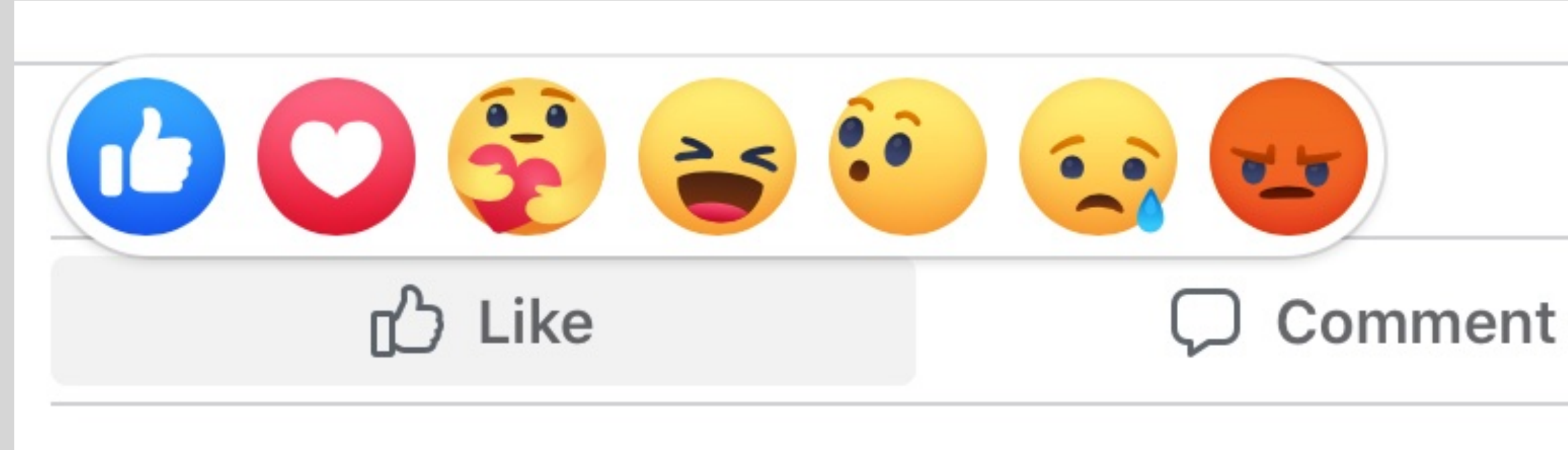
**concept** Recommending

**purpose** use prior likes to recommend

**principle** user's likes lead to ranking of kinds of items, determining which items are recommended



# what's going on in Facebook reactions?



Which concept is at play here?  
Upvoting, Reacting, Recommending?



YRYF5Y

<https://class.mit-sdg.dev/join>

**concept** Upvoting

**purpose** rank items by popularity

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**concept** Reacting

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**concept** Recommending

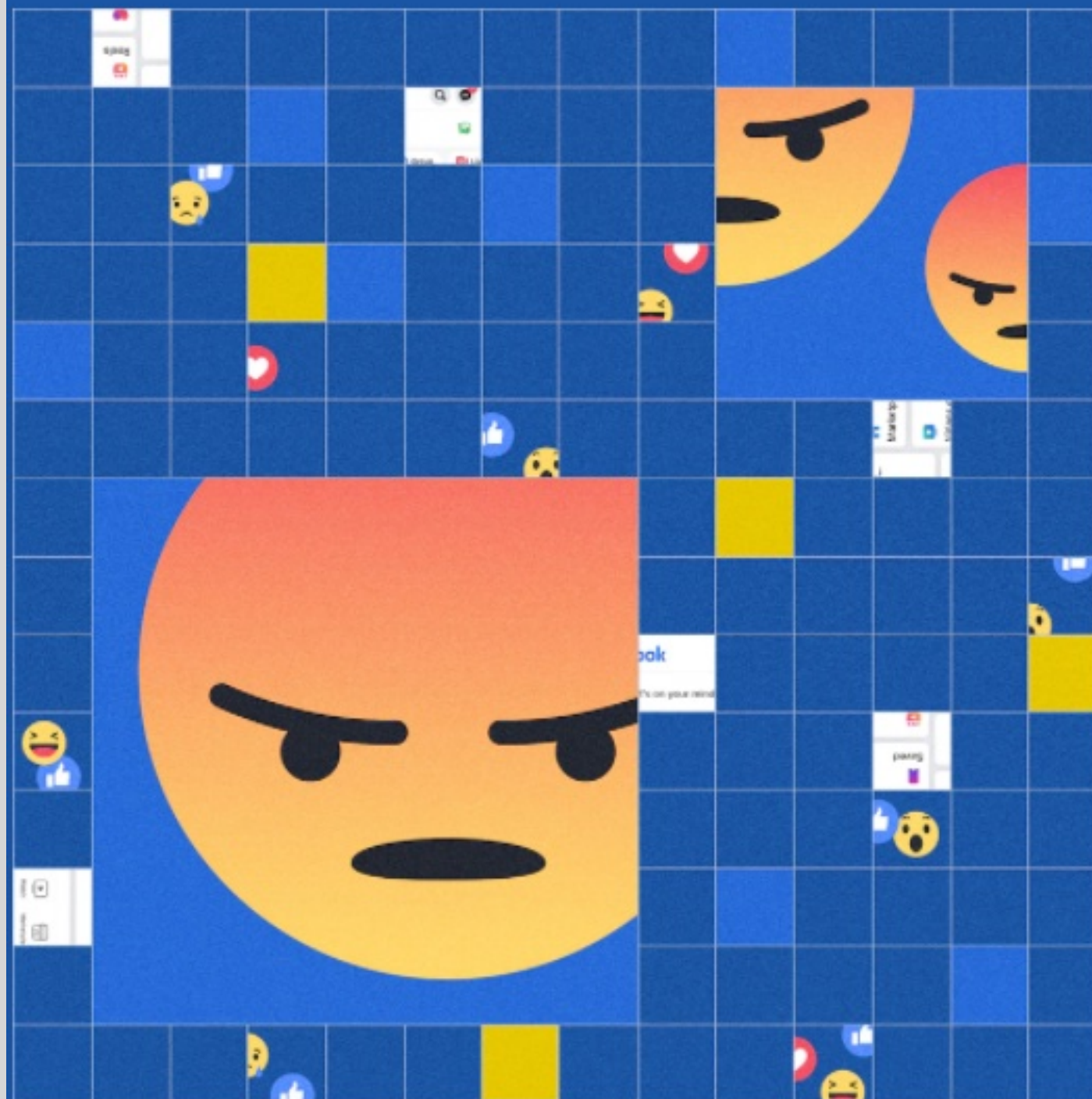
**purpose** use prior likes to recommend

**principle** user's likes lead to ranking of kinds of items, determining which items are recommended

## Facebook under fire

# Five points for anger, one for a 'like': How Facebook's formula fostered rage and misinformation

Facebook engineers gave extra value to emoji reactions, including 'angry,' pushing more emotional and provocative content into users' news feeds



(Washington Post illustration; Facebook screenshots; iStock)

# Emoji reactions were a cute addition to Facebook. They became a headache.

Facebook's algorithms gave an "angry" reaction five times the weight of a traditional "like." After years of research, employees realized the result was more "toxicity."

this was true, but in fact  
all emojis were equally weighted



**Andy Ostroy** ✓

@AndyOstroy



Seems the only #Wall @realDonaldTrump's built is the one between him and @FLOTUS #Melania #trump



♥ 8,221 8:15 PM - May 2, 2017

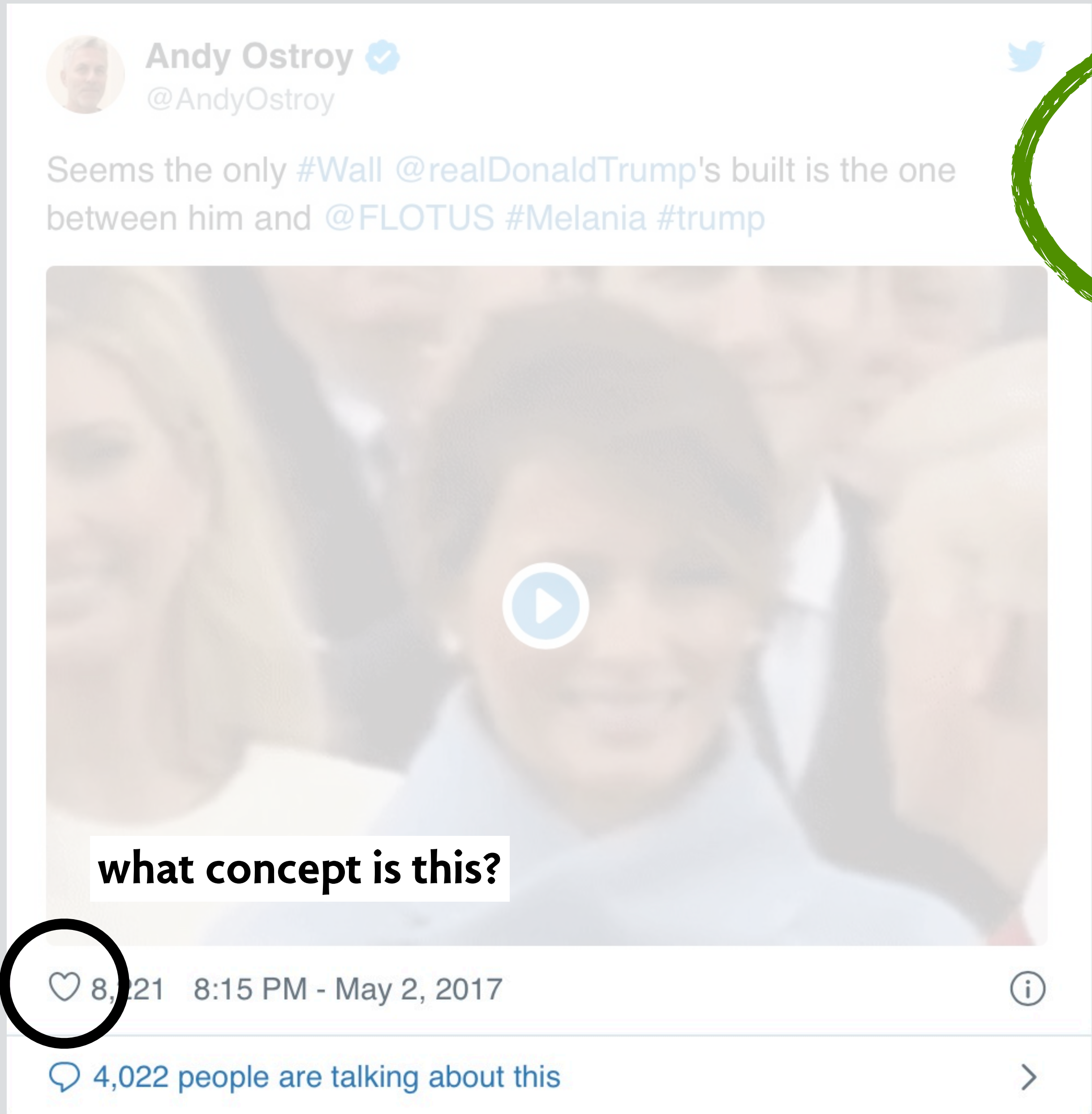


💬 4,022 people are talking about this



**MELANIA TRUMP** liked your Tweet

Seems the only #Wall @realDonaldTrump's built is the one between him and @FLOTUS #Melania #trump [pic.twitter.com/XiNd2jiLUF](https://pic.twitter.com/XiNd2jiLUF)



**what concept is this?**



**concept** Reacting  
**purpose** share reactions to an item



**concept** Bookmarking  
**purpose** save items to revisit later



**The Boston Globe**  @BostonGlobe · 21h

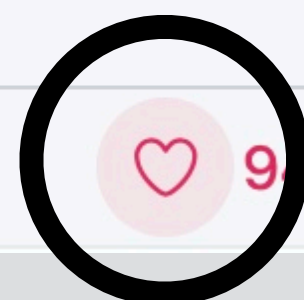
Andrew Yang would fine gunmakers for deaths caused by their products.



Yang would fine gunmakers for deaths caused by their products - Th...  
You probably know Andrew Yang wants to give every American \$1,000  
a month. Something you might not know: He wants to fine gun ...  
[bostonglobe.com](https://www.bostonglobe.com)

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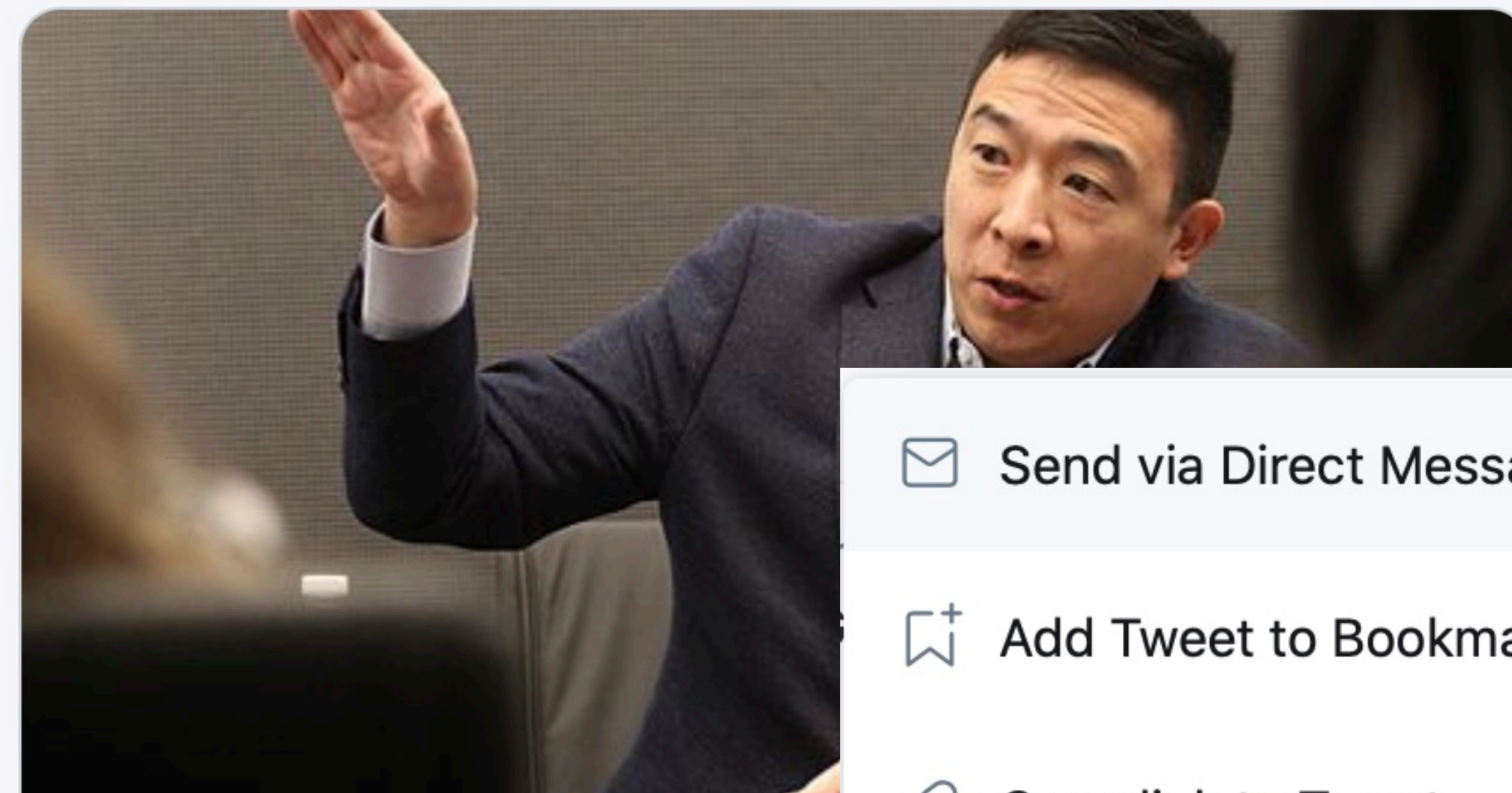


**concept** Reacting  
**purpose** share reactions to an item



**The Boston Globe**  @BostonGlobe · 21h

Andrew Yang would fine gunmakers for deaths caused by their products.

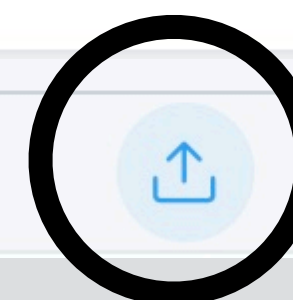


Yang would fine gunmakers for deaths  
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a month. Something you might not know  
[bostonglobe.com](https://www.bostonglobe.com)

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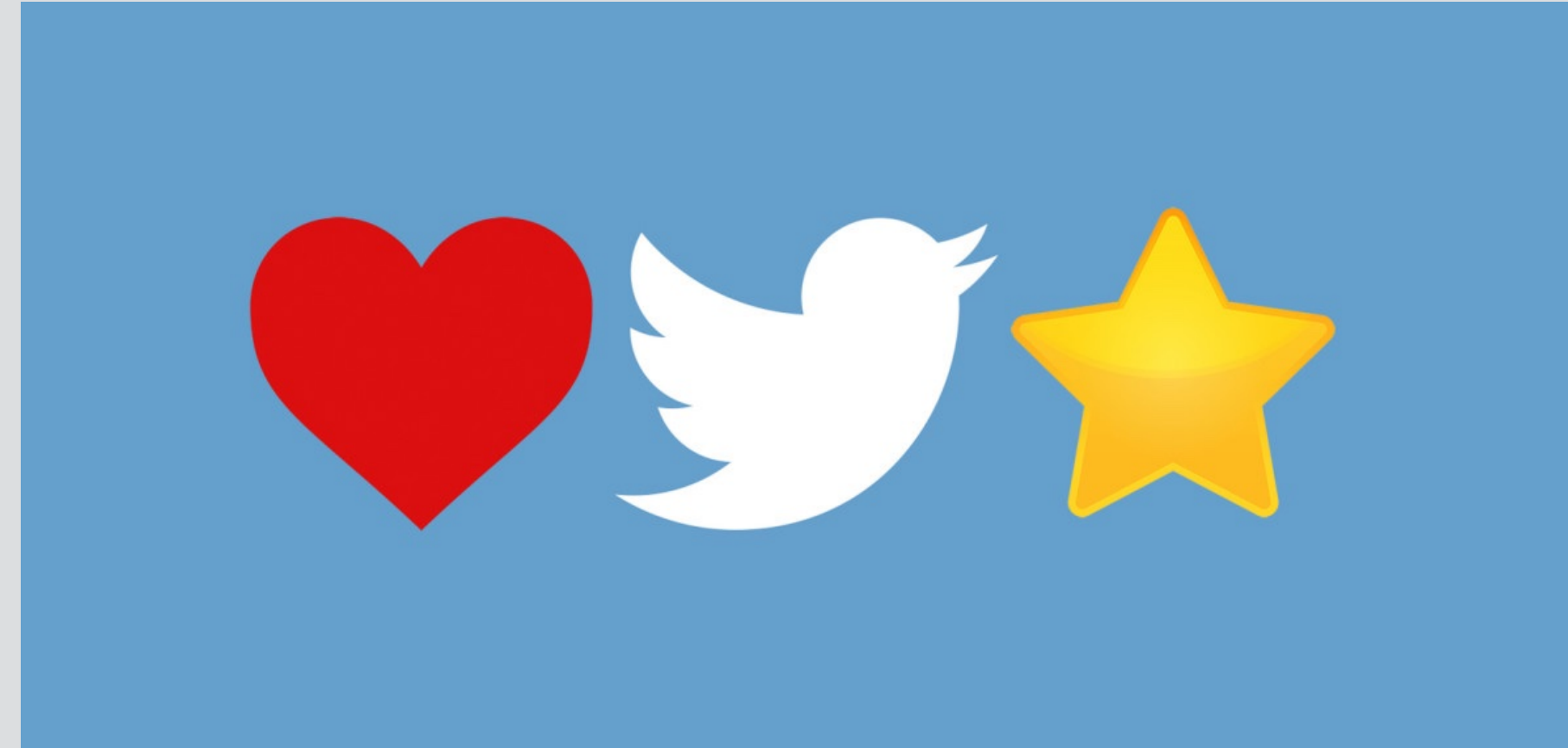
 94



**concept** Bookmarking  
**purpose** save items to revisit later

-  Send via Direct Message
-  Add Tweet to Bookmarks
-  Copy link to Tweet
-  Share Tweet via ...

# discussion: why did twitter say this?



Nov 2, 2015: Twitter changes Favorite (Star) to Like (Heart)

We are changing our star icon for favorites to a heart and we'll be calling them likes... **We know that at times the star could be confusing, especially to newcomers.** You might like a lot of things, but not everything can be your favorite. *Twitter press release*

concepts are  
inventions

# zoom's key concept

**concept** SelfJoining

**purpose** let participants in an event join by themselves; prevents host needing to add one by one

**principle** A host creates a link to an event and shares it; a participant who receives it can then join the event using the link

# tracing zoom's self-joining concept



## Skype

initially P2P (2003)  
Microsoft (2013)



## FaceTime

Apple (2010)



## Google Hangouts

in Google+ (2011)  
own product (2013)  
Duo replaces (2016)



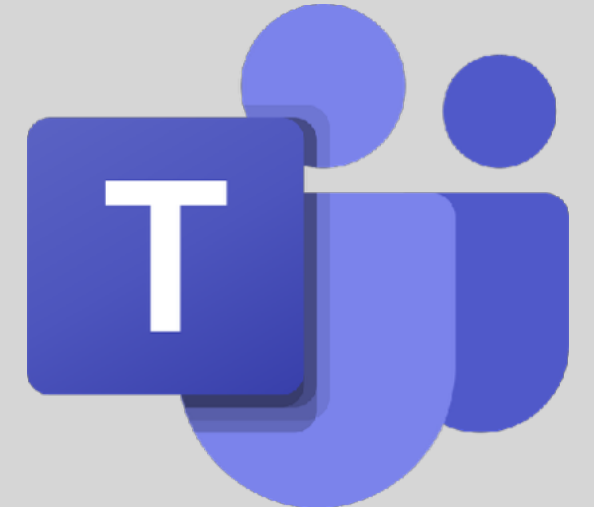
## Zoom

Eric Yuan (2013)



## Google Meet

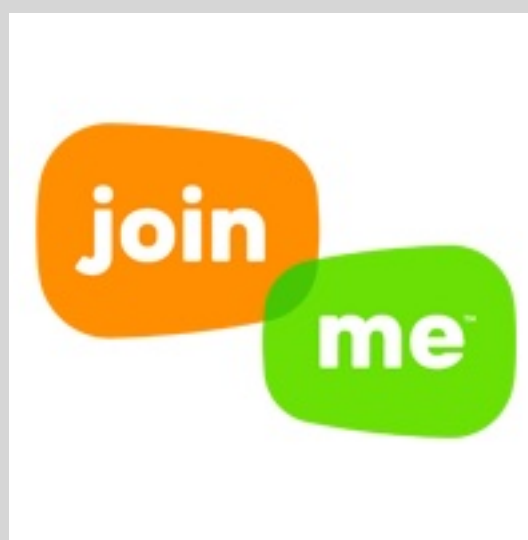
launched (2017)  
absorbs Duo (2022)



## Microsoft Teams

launched (2017)

SelfJoining  
added to Skype  
April 2020



## Join.me

LogMeIn (2010)

SelfJoining  
adopted by Zoom

SelfJoining  
added to Teams  
c. June 2022

## Schindler's PORT elevator



**concept** DestinationDispatching

**purpose** improve elevator utilization; prevent needless stops when some elevators are idle

**principle** A rider selects a destination floor and is told to wait for an elevator at a given bank; they take the first car that arrives and it will eventually stop at their floor.

a real change  
demanded in user  
behavior

## Schindler's PORT elevator



### A boost to traffic performance

Schindler PORT groups passengers by destination to provide the shortest possible trip for every rider, avoiding chaotic elevator runs and random, multiple stops.

### User-friendly operation

With Schindler PORT's universal card reader, a rider simply swipes a personal access card at the Schindler PORT terminal for an elevator car to be immediately assigned with an approved destination.

### Excellent personalized service

Schindler PORT can customize your journey, whether it be a special VIP trip, a streamlining patient transport or allowing for more approach time, longer door opening time or additional space for passengers with special needs.

### Enhanced building security

Floor access is defined by the building management through access cards or smartphones. Schindler PORT can also be integrated with turnstiles to further enhance building security.



Schindler Miconic 10 (1992)  
first commercial implementation



Leo Port, electrical engineer in Sydney  
patents destination dispatch (1961)  
but lets it expire (1977)



Schindler introduces PORT (2009)  
“Personal Occupant Requirement Terminal”

# mutual matching

**concept** MutualMatching

**purpose** ... prevents ...

**principle** Two people are introduced to each other as a potential match; each of the two can either accept or reject the match; if both accept, the match succeeds.

**Speed dating (1998)**  
invented by Rabbi Yaacov Deyo

7-8 minutes dates  
marked yes/no on a card after  
concept went viral quickly



# mutual matching

**concept** MutualMatching

**purpose** let two people decide if they match without knowledge of the other's decision; prevents humiliation of rejection

**principle** Two people are introduced to each other as a potential match; each of the two can either accept or reject the match; if both accept, the match succeeds.

# reciprocal reviewing

**concept** ReciprocalReviewing

**purpose** let parties who interacted review each other independently; prevents pressure or retaliation

**principle** After an interaction between two parties, they are invited to review each other; the reviews are not visible until both have reviewed or some deadline has passed.

**Rate your experience with  
Lorraine & Greg**



Terrible



Bad



Okay



Good



Great

You just checked out of Lorraine & Greg's place. Now, take a minute to reflect on the experience and share a quick review.

You'll have space to leave private feedback, just for Lorraine & Greg, and public comments for future guests too.

We won't share any of your responses until after Lorraine & Greg leaves feedback too.

Airbnb had mutual reviews on launch in 2008  
Updated with blinding in 2014  
Ratings dropped, more for low quality listings  
Review participation increased

concepts that  
made companies

# core concepts & the companies they made



Zoom  
**SelfJoining**



Calendly  
**SelfScheduling**



Tinder  
**MutualMatching**



Snapchat  
**Expiring**



Pinterest  
**Pinning**



Dropbox  
**Syncing**



Adobe  
**Layering**



Google  
**PageRanking**

spotting  
concept flaws

# spotting concept flaws

## **concept** ChoreAssigning

**purpose** allocate chores fairly among living group members; prevent uneven burden

**principle** Chores are assigned to different members and marked as complete when they are done.

## **concept** Preregistering

**purpose** collect information about demand for classes for resource planning and prioritize most enthusiastic students with waitlists.

**principle** Before the preregistration deadline, students preregister their intended class selections, and can then add and remove selections prior to the registration deadline.

## **concept** Playlisting

**purpose** let listeners select songs and then play them in the selected order.

**principle** Playlist consists of an ordered sequence of songs dictating the order of play, starting at any point.

purpose doesn't explain why

purpose is actually two purposes

principle is not a course of action

principle does not fulfill purpose

parting  
thought

“Well, the most important thing that was new was the idea of URI — or URL.”

Tim Berners Lee (Wired, 1999)

Why was the URL the key concept in the invention of the web?

And why does TBL now think that this invention is threatened?